

Software Design II - Concurrent System Design SOFT ENG 3BB4

Term 1, Fall 2024

Course Information

Instructor: Prof. Ryszard Janicki

- Office: ITB 217
- Phone: 525-7070 ext: 23919
- Office hours: Monday 1:30 – 2:30 pm
- Email: janicki AT mcmaster.ca

Teaching Assistants:

- Andrew Mitchel: mitchaj2@mcmaster.ca
- Atiyeh Sayadi: atiyeh.sayadi@yahoo.com
- Jatin Chowdhary: chowdhaj@mcmaster.ca
- Heywon Jo: joh6@mcmaster.ca

Office Hours:

- TBA

Course website: <https://www.cas.mcmaster.ca/~se3bb4>

Lectures:

Monday	02:30 PM - 03:20 PM	ITB AB103
Wednesday	02:30 PM - 03:20 PM	MDCL 1102
Friday	04:30 PM - 05:20 PM	ITB AB102



Tutorials:

Wednesday	12:30 - 02:20 PM	ABB B118	Start September 11. 2024
Thursday	12:30 - 02:20 PM	T13 107	Start September 11. 2024
Friday	08:30 - 10:30 AM	BSB 117	Start September 11. 2024



Submissions will be via Avenue.

Course Outline (Tentative)

- Nature of Concurrency. Processes and Threads. Concurrent Execution. Labelled Transition Systems. Shared Objects and Mutual Exclusion. Monitors and Condition Synchronization. Deadlock. Safety and Liveness Properties. Model Based Design. Dynamic Systems. Message Passing. Concurrent Architectures. Timed Systems. Program Verification. Logical Properties. Petri Nets and other models.

Texts:

- J. Magee, J. Kramer, *Concurrency: State Models & Java Programming*, 2nd Edition, J. Wiley 2006.

The course may not always follow the text-book closely.

Lecture Notes (copies of transparencies) will be on the website a few days after a class.

Software Tools: You will have to obtain the LTSA tool from <http://www.doc.ic.ac.uk/~jnm/book>

Calendar Description:

- **Processes, threads, concurrency; synchronization mechanisms, resource management and sharing; objects and concurrency; design, architecture and testing of concurrent systems.**

Mission:

- **The mission of this course is to give students an understanding of the basic techniques of concurrent system design and to introduce students to the appropriate modelling and analysis techniques.**

Learning Objectives: Preconditions:

- Students should have basic knowledge of discrete mathematics (especially of logic, sets and relations), basic knowledge of data structures and algorithms. They should be familiar with principal ideas of software development and know how to program in at least one programming language.

Learning Objectives: Postconditions

1. **Students should know and understand**
 - a. **Basic models of concurrency, both algebraic and graphical**

- b. The concepts of process and thread
- c. Resource management and sharing
- d. Basic ideas of design and architecture of concurrent systems
- e. Problems with testing and verification of concurrent systems

2. Students should be able to

- a. Specify concurrent systems with process algebras and Petri nets.
- b. Implement concurrent systems with processes, threads and other available tools.
- c. Verify and test crucial parts of concurrent systems.

Evaluation

Assignments	30%	Three assignments (3 x 10=30%).
Midterm test	20%	(60 minutes midterm test plus 20 minutes for technology) (20%, take home, virtual on Avenue)
Final examination	50%	There will be a 2.5 hours (one double side cheat sheet will be allowed) final examination
Total	100%	

Lecture notes and submissions will be via Avenue.

Late assignments will not be accepted because solutions will be posted on the website a day after the due day. Although you may discuss the general concept of the course material with your classmates, your assignment must be your individual effort.

Midterm:

- TBA

MSAF policy:

For each MSAF the weight will be moved to the final exam. More information on the MSAF procedure can be found at this [link](#).

Important

The instructor and university reserve the right to modify elements of the course during the term. The university may change the dates and deadlines for any or all courses in extreme circumstances. If either type of modification becomes necessary, reasonable notice and communication with the students will be given with explanation and the opportunity to comment on changes. It is the responsibility of the student to check their McMaster email and course websites weekly during the term and to note any changes.

Academic Dishonesty

Academic dishonesty consists of misrepresentation by deception or by other fraudulent means and can result in serious consequences, e.g. the grade of zero on an assignment, loss of credit with a notation on the transcript (notation reads: A Grade of F assigned for academic dishonesty@), and/or suspension or expulsion from the university.

It is your responsibility to understand what constitutes academic dishonesty. For information on the various kinds of academic dishonesty please refer to the Academic Integrity Policy, specifically Appendix 3, located

at http://www.mcmaster.ca/senate/academic/ac_integrity.htm

The following illustrates only three forms of academic dishonesty:

1. Plagiarism, e.g. the submission of work that is not one's own or for which other credit has been obtained.
2. Improper collaboration in group work.
3. Copying and using unauthorized aids in tests and examinations.

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